

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester IV Examination, 2022-23

Paper: BSC-04A — Elements of Geology - II

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 058432

*Instructions: Answer **five** questions in all, selecting at least **two** questions from each of Sections A and B. Section-C is compulsory. All questions carry equal marks.*

SECTION — A

Question 1. What is the significance of the geological time scale? Explain the representative life forms of different geological periods. [14]

Question 2. Answer the following:

- (a) What are the different stratigraphic units? Describe the difference between lithostratigraphic and chronostratigraphic units. [7]
- (b) Write an essay on the principles of stratigraphy. How are these principles helpful in establishing a geological record of the strata? [7]

Question 3. Answer the following:

- (a) Discuss the geological differences between the Andaman and Lakshadweep Islands of India. [5]
- (b) Discuss briefly the desertification of Thar and mention the characteristic topography of deserts. [5]
- (c) Write a short note on the mass extinction event related to dinosaurs. [4]

Question 4. Answer the following:

- (a) Discuss the difference between the eastern and western coastal plains of India. [4]
- (b) Discuss “Present is the key to the past”. [4]
- (c) Name the important peninsular plateaus of India. [3]
- (d) Write a short note on the “Law of lateral continuity”. [3]

SECTION — B

Question 5. Discuss the conditions for fossilization of organisms and various kinds/types of fossils. [14]

Question 6. Define paleontology. Describe the scope of paleontology in relation to other sub-disciplines of geology. [14]

Question 7. Write explanatory notes on the following:

- (a) Siwalik vertebrate fauna. [6]
- (b) Traces of biological activity. [4]
- (c) Living fossil. [4]

SECTION — C

Question 8. Define the following in a maximum of five sentences each:

- (i) Acme Zone [2]
- (ii) Lagoon [2]
- (iii) Barren Island [2]
- (iv) Carbonization [2]
- (v) Mould and Cast [2]
- (vi) Pseudo fossil [2]
- (vii) Facies fossil [2]